ML Workflow

- Data gathering
- Data Cleaning → Identify & Rectify any errors, validity issues, inconsistencies & inaccuracies in the data.
- EDA → Identify patterns & trends in the data i.e. understand the data, using which we identify which features are relevant/irrelevant to the model & take appropriate actions.
- Feature Engineering → Create our own features based on the available features in an attempt to improve the accuracy of the model
  → Experimental
- Data Preprocessing → Most data sets we use are not in a form suitable to train our model. We need to convert the data in a form for training the model.
- Model training
- Model tuning
- Deployment
- Monitor

AWS Sagemaker — An AWS that allows you to implement the complete ML lifecycle in one place. Usually for each stage of the ML workflow we are using different tools.

AWS Sagemaker

- Model Training
- Model Tuning
- Model Monitoring
- Deployment
- Feature Engineering

Advantages

- it's a fully managed service
- Provides wide range of libraries for ML workflows
- Provides " " " algorithms for various problems
- One-click deployment
- Integration with other AWS services
- Provides notebooks & IDE for development
- Detailed guides & tutorials
- Huge Community & online support

X

AWS IAM → Deals with permissions & access

IAM → Identity & Access Management

- Root User
- IAM Group
- IAM User
- IAM Role

X

AWS EC2

Virtual computer on the cloud, where we can run our tasks / computations & we can also scale the requirements according to our needs.

Elastic Cloud Compute

EC2 Instance

X

AWS S3

Virtual container on the cloud, where we can store & save our files. i.e. online storage service.

S3 Bucket

Simple storage service

X

SDK

Collection of tools that helps with creating a software completely end-to-end.

Software Development Kit

Code

APIs

Documentation

Libraries

Guides

Advantages

- Efficient Development
- Reduced time complexity
- Seamless Integration
- Reduced costs

- Using Python to train models
- Need a way for Python to interact with AWS services
- Use AWS SDK for python to create, configure & manage AWS services.

Python $\xrightarrow{\text{SDK}}$ AWS

|Boto3| → AWS SDK (Python)

AWS

S3 Bucket

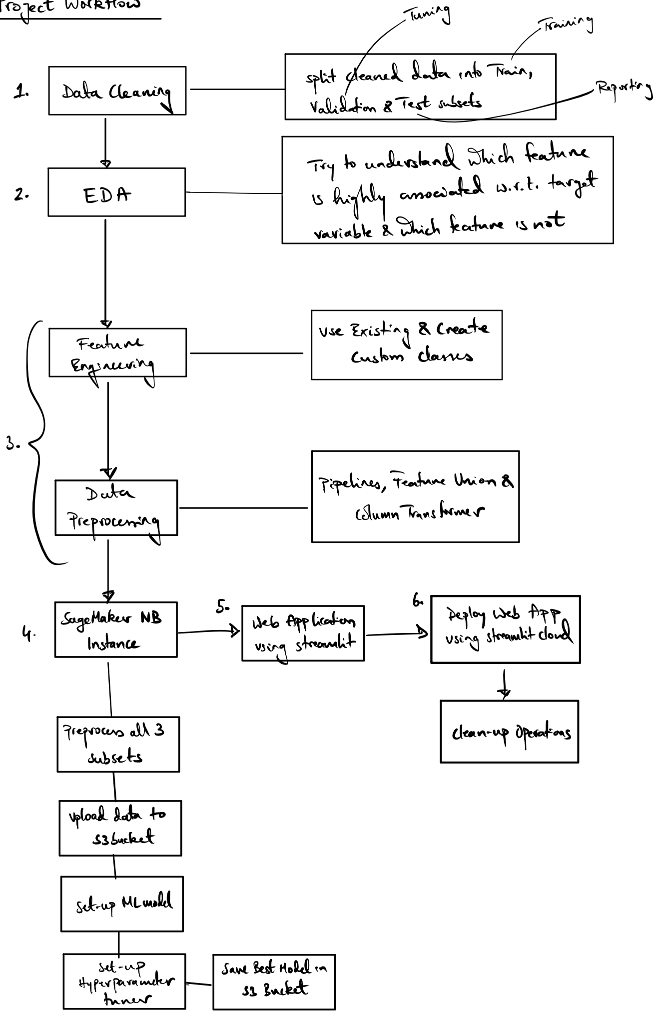
IAM Role

EC2

SageMaker

X

Project Workflow



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Data Cleaning

- Functional programming
  - → readable
  - → clear
  - → robust
- Follow Pandas best practices

Things to Do	Things Not to Do
- Check data types of features	- Learn any patterns from the dataset → "Snooping Bias" or "Data leakage"
- Check validity of values for each feature	- plots
- Check accuracy of values for each feature (extreme values)	
- Check for duplicates	

Flow

Read Dataset → Validate Data Types → Check Duplicates → Validate Each Feature → Document observation → Implementing cleaning operations → Split Data into subsets → Export Data

- Implementing cleaning operations: Functional Programming
- Split Data into subsets: Train, Validation, Test
- Export Data: .csv files

- Train (Learning)
- Val (Revision)
- Test (Final)

- Tune the parameters on the validation set → Best parameters

Re-train the model
- Best parameters
- Train + Val set

EDA

- Exploring the dataset, by utilizing visualization tools & stastical measures to understand & extract underlying patterns & information within the data.

- gives a clearer picture of the data, helps us make data-informed decisions & solve crucial business problems with much fewer assumptions & more facts

- Look at & analyze numbers & plots

- Descriptive stats
  - Central tendency
  - Dispersion/spread
  - Distribution of Variables
  - Symmetry & Shape of variables

  } summarize data

- Inferential stats
  - Strength of Association (correlation)
  - Hypothesis Testing

  } sample → population

- plots/Graphs
  - Univariate
  - Bivariate
  - Multivariate

- Descriptive stats
  - Measure of location
    - Mean
    - Mode
    - Median
    - Percentiles
    - Quartiles

    } 1st moment of distribution
  - Measure of spread
    - Variance
    - Standard Deviation
    - Coefficient of Variance
    - Mean Absolute Error (MAE)

      } outliers
    - Inter Quartile Range (IQR)
    - Median Absolute Deviation from Median (MAD)

      } Robust

    } 2nd movement of distribution
  - Measure of symmetry — Skewness } 3rd movement of distribution
  - Kurtosis } 4th movement of distribution

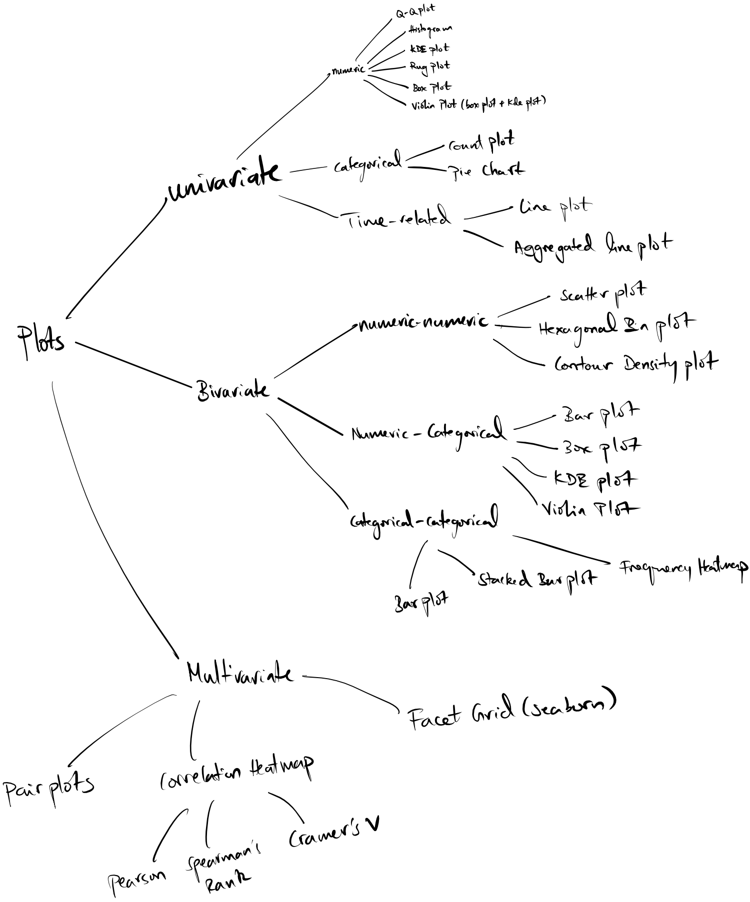
- Strength of Association
  - Pearson's Correlation — b/w 2 numeric variables assuming they are linearly related
    - Correlation — metric that helps us understand the Relation b/w two variables
  - Spearman's Rank Correlation — doesn't make linear assumption
    - monotonic relationships
  - Cramer's V — 2 categorical variables
  - numeric vs categorical ??

- inferential stats
  - Hypothesis testing
    - Test for normality
      - Shapiro-wilk Test
      - Anderson-Darling Test
    - Test for Association
      - Numeric Variables
        - Pearson's Test
        - Spearman's Test
      - Categorical Variables — Chi-D Test
      - Numeric-categorical Variable
        - one-way ANOVA Test
        - Kruskal-Wallis Test

Steps Involved

- State the Hypotheses
  - Null Hypotheses (H_0)
  - Alternate Hypothesis (H_a)
- Determine significance level (alpha 5%)
- Determine which test to perform
- Collect necessary data (sample)
- Obtain critical values → (based on tables, dof etc)
- Compute Test statistic (& p-value) → (based on sample data)
- Compare
  - significance lvl vs p-value

    or
  - Critical values vs Test static
- State conclusions

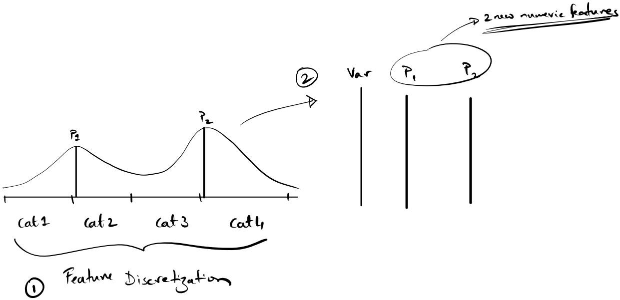


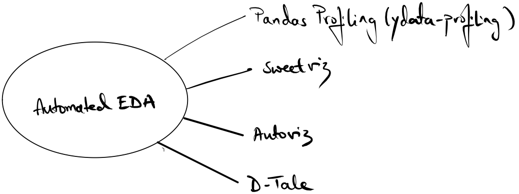
Plots
univariate
numeric
Q-Q plot
Histogram
KDE plot
Rug plot
Box plot
Violin Plot (box plot + kde plot)
categorical
Count plot
Pie Chart
Time-related
Line plot
Aggregated line plot
Bivariate
numeric-numeric
Scatter plot
Hexagonal Bin plot
Contour Density plot
Numeric - Categorical
Bar plot
Box plot
KDE plot
Violin Plot
categorical-categorical
Bar plot
Stacked Bar plot
Frequency Heatmap
Multivariate
Facet Grid (seaborn)
Pair plots
Correlation Heatmap
Pearson
Spearman's Rank
Cramer's V

Sequence of steps

- Import libraries
- Read the training data subset
  - Check data types
  - Fix data types (if applicable)
- Gather high-level summary of the data
  - .info() method
  - .describe() method on numeric & categorical features
- High-level analysis of missing values
  - Bar plot
  - Count plot
  - Missingno
- High-level analysis of outliers:
  - Isolation forest
- Correlation Analysis (heatmaps)
  - Numeric (Pearson's / Spearman's)
  - Categorical (Cramer's V)
- Detailed Analysis of each Feature
  - Summary
  - Univariate plots
  - Bivariate plots (w.r.t. the target variable)
  - Hypothesis Testing (normality, strength of association)
  - Multivariate plots
  - Inspect missing values & extreme values in-depth
    - Filter for necessary subsets
    - inspect values of other features (plots, summary stats)
  - Note observations

- Feature Engineering
    - Create new features
    - Repeat above steps for newly created features
- Repeat above steps iteratively





Feature engineering

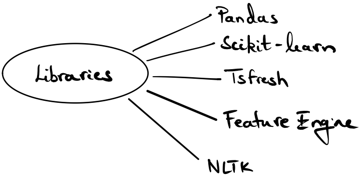
- process of creating new features from the existing features in the dataset.

- aim is to try & come up with new features that can help boost the model performance.

- experimental step; creativity

Data Pre-processing

- process of converting the dataset into a form suitable for training a model
- Almost all features require some sort of pre-processing b4 model training.



Numeric variables

1. Extreme Values

Ask:

- Are the extreme values genuine?
- Are the extreme values erroneous?

Identify:

- Mean & std. Dev.
- IQR
- Beyond / below percentiles
- Isolation forest
- Plots
  - Box plot
  - KDE plot + Rug Plot

Remedial steps:

- Delete extreme values
- Cap/Floor extreme values
  - Arbitrary values
  - Mean +/- (3 * std. Dev.)
  - Q1-1.5 * IQR, Q3 + 1.5 * IQR
  - 5th & 95th percentiles
- Represent extreme values as missing (for later imputation)

2. Missing Values

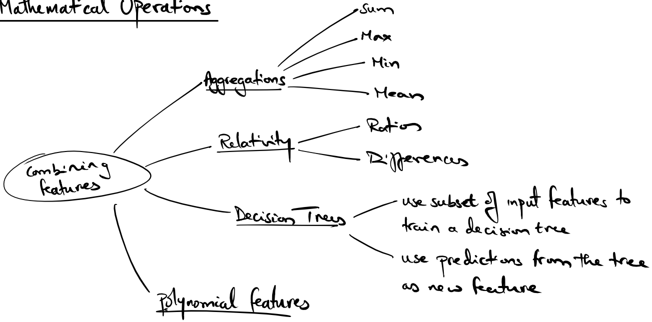
Ask:

- Are values missing b/c they don't exist?
- " " " " " weren't recorded?

Remedial steps:

- Delete missing values
- Create indicator columns
- Impute missing values
    - Mean
    - Mode
    - Median
    - Constant Value
    - Algos (MICE, KNN)

3. Mathematical Operations



- Functions
  - Log
  - Square Root
  - Reciprocal
  - Exponential

- Transformations
  - Power Transformations
  - Box-Cox Transformation
  - Yeo-Johnson Transformation

- Feature Discretization
  - Equal width
  - Equal Frequency
  - Arbitrary Intervals
  - K-Means Clustering

4. Feature Scaling

- Standardization
- Normalization
- Median & IQR

Categorical Variables

1. Missing Values

- Most common value
- Arbitrary value

2. Manipulating Categories

- Group Rare Categories
- Replace related categories with meaningful values
- Convert to binary Categories

3. Encoding Categories

- Ordinal encoding
- one-hot encoding
- Rare-label encoding
- Frequency encoding
- Target mean encoding

Date & Time Variables

1. Extracting Features from Dates

- Day
- Month
- Year
- Quarter
- Weekday
- Day of the Year
- Day of the Week
- Year Half (from Quarter)

2. Extracting Features from Timestamps

- Hour
- Minute
- Second

3. Creating new Features

- Difference b/w 2 features time/date
- Represent "time lag" in minutes, seconds

Text Variables

1. Aggregations

- Frequency of characters
- " " words
- " " unique words

2. Ratios

- Take ratios from above created features

Feature Selection

1. Filter Methods

- Involves ranking each variable in the dataset & then selection
- Variables are ranked based on some statistical metric calculated
  - Chi-square test statistic
  - F-test statistic
  - Correlation Coefficient
  - Mutual information
- These techniques don't involve any Machine Learning flavours
- " " are fast & easily scalable.

2. Wrapper Methods

- These techniques involve generating various subsets of the dataset
- Each subset is evaluated against a ML model
  - involves training multiple models
  - Trains one model for each subset
- The variables present in the best performing subset are the selected features
- Algos
  - Exhaustive Search
  - Forward Elimination
  - Backward "

3. Embedded Methods

- The selection mechanism used, is built-in into the ML model as part of its training phase
  - linear regression
  - lasso Regression
  - logistic "
  - linear models
  - Decision Trees
  - Random Forest
  - Tree-based models
- These techniques involve training only a <u>single model</u>
- Features are selected by ranking based on estimated coefficients or feature importances produced by the trained model.

Major Components

- Built-in Transformers
- Custom Transformers
    - Python class
    - Python function

scikit-learn compatible transformer

- Function Transformer
- Feature Union
- Pipeline
- Column Transformer

RBF kernel
- function
- Calculates similarity b/w two values

Input - x
Ref - y

$$\boxed{e^{-(\gamma \|x-y\|_2^2)} \longrightarrow e^{-(\text{gamma} \times \text{distance}^2)}}$$